



Viscosity of Newtonian Fluids

Group # 2

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Introduction

- **Purpose of lab**
 - To analyze the relationship between capillary size, pressure drop and flow rate
 - Obtain “m” and “n” viscosity parameters



Theory

- Underlying physical principles, governing equations
 - Pressure Change
 - Viscosity
 - Non-Newtonian Fluid Properties
- Hagen Poiseuille Equation and Ostwald-deWaele follow, respectively:

$$Q = \frac{\Delta P \pi R^4}{8 \mu L}$$



$$\tau_{yx} = \pm m \left| \frac{dv_x}{dy} \right|^n$$

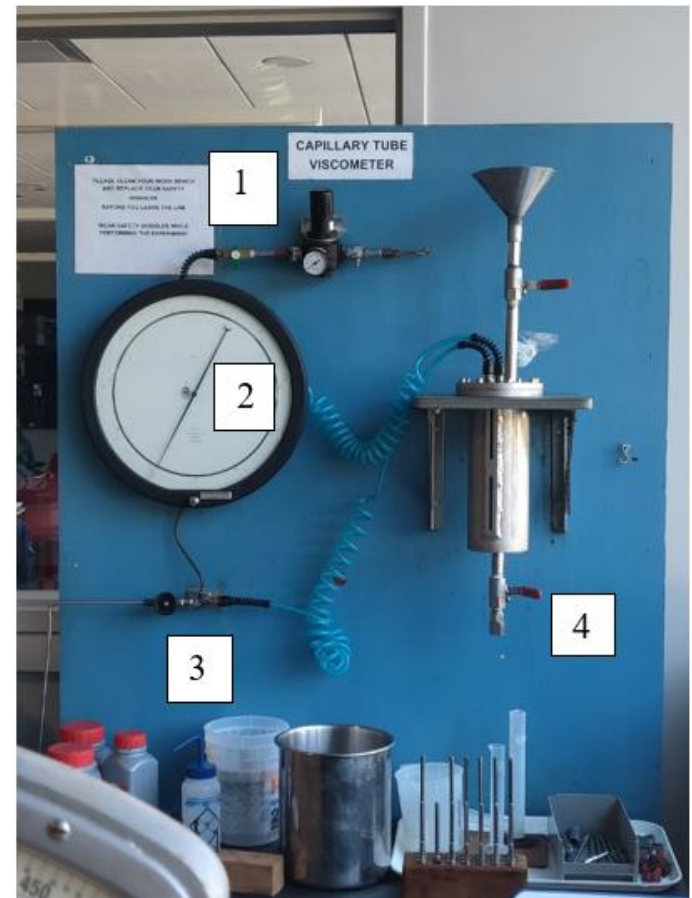


Theory

- Assumptions/limitations
 - Temperature doesn't affect viscosity in this lab
 - All equipment was calibrated correctly
- Connection to experiment
 - Experiments were done to analyze the viscosity of the Carbopol solution with varying pressure drops and capillary tubes

Apparatus

- 1: Regulator
- 2: Test Pressure Gauge
- 3: Pressure Release Valve
- 4: Pressurization Vessel
- 5: Drain Valve (not pictured)



Material & Supplies

- Carbopol Polymers Powder
- Warm Distilled Water
- 1 N NaOH Liquid Solution
- pH Paper
- Scale
- Capillary Tubes
- Mixing Apparatus
- 2 L Plastic Container
- Stopwatch

Procedure

- Make the Carbopol solution the day of the pre-lab
 - Make sure to add NaOH to get the pH of the solution to be between 5-6
- Fill in the vessel with all your solution and measure the amount of solution you collected by weight at a certain time.
 - It's easier to calculate the flow rate by measuring the density of the solution
 - Make sure to release the pressure after each trial run of each tube.
- Make sure to discard the solution into a Ziploc bag after the experiment is over.

Experimental Challenges



- Things to watch out for
 - Adjust the pressure of the valve from the back if the pressure rises too quickly or too slowly
 - Getting the pressure to be at the exact point that you intended it to get it to
- Lessons Learned
 - Depressurize the vessel after capillary tube
 - Don't have to keep cleaning the cylinder after each measurement.
 - Just subtract the weights to get the amount collected for the run

Safety

- Mix the Carbopol solution carefully
 - (solution can cause burns and irritation from Carbopol powder)
- Watch the pressure throughout the lab and depressurize often
 - (Maximum pressure: 60 psig)

Comment Summary

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1. 5/5
2. 94/100

Page 2

3. Technical details?
4. 12/15

Page 3

5. Not used for analysis
6. Show the other equation used to get the m and n parameters
 - i. Don't paste images of equations
7. 12/15

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